

An RFD Operator Algebra with Non-RFD Maximal C^* -Cover

An Exact Literature Counterexample to arXiv:2107.14275

Status note for arXiv:2107.14275

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Status: literature already answered; full negative answer. Hartz's 2022 construction directly disproves the question. This packet records the identification and proof chain and claims no original result.

1 Source question

An operator algebra is *residually finite dimensional* (RFD) when its matrix norms are separated by finite-dimensional completely contractive representations. Thompson asks whether this property must pass to the maximal C^* -cover [1, Question 1]:

$$\mathcal{A} \text{ RFD} \quad \xRightarrow{?} \quad C_{\max}^*(\mathcal{A}) \text{ RFD}.$$

The maximal C^* -cover of an operator algebra $\mathcal{A}$, denoted $(C_{\max}^*(\mathcal{A}), \mu)$, satisfies the universal property that every representation of $\mathcal{A}$ lifts to a unique $*$ -representation of $C_{\max}^*(\mathcal{A})$. Hence, the representations of $\mathcal{A}$ are in one-to-one correspondence with the $*$ -representations of $C_{\max}^*(\mathcal{A})$. As residual finite-dimensionality is encoded in the structural properties of spaces of representations, one could ask whether residual finite-dimensionality of operator algebras is closely linked to that of their maximal C^* -covers. This provides us with our main inquiry:

Question 1. If $\mathcal{A}$ is an RFD operator algebra, then is $C_{\max}^*(\mathcal{A})$ necessarily RFD?

There have been several instances for which Question 1 has been answered in the affirmative [9], [11]. Most notably, it was shown that $C_{\max}^*(\mathcal{A})$ is RFD whenever $\mathcal{A}$ is finite-dimensional. The maximal C^* -cover is also RFD for Popescu's disk algebra, the Schur-Agler class of functions, as well as some semigroup algebras and spaces of analytic functions. We remark that Question 1 is also related to a result of Pestov [30, Theorem 4.1].

Now we outline the structure of the paper and our approach to the main question.

Figure 1: The source question in arXiv:2107.14275, PDF p. 2.

2 Answer and counterexample

Answer. No.

Let $A(\mathbb{D})$ denote the disc algebra, regarded as a subalgebra of $C(\mathbb{T})$, and define

$$\mathcal{B} = \left\{ \begin{bmatrix} f & 0 \\ h & \bar{g} \end{bmatrix} : f, g \in A(\mathbb{D}), h \in C(\mathbb{T}) \right\} \subset M_2(C(\mathbb{T})). \quad (1)$$

Hartz proves that $\mathcal{B}$ is a unital RFD operator algebra and, more strongly, is completely isometrically 2-subhomogeneous [2, Theorem 1.3].

be the disc algebra. We think of $A(\mathbb{D})$ as a subalgebra of $C(\mathbb{T})$ by virtue of the maximum modulus principle.

Theorem 1.3. *Let*

$$\mathcal{B} = \left\{ \begin{bmatrix} f & 0 \\ h & \bar{g} \end{bmatrix} : f, g \in A(\mathbb{D}), h \in C(\mathbb{T}) \right\} \subset M_2(C(\mathbb{T})).$$

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Then:

- (a) $\mathcal{B}$ is a unital operator algebra that is RFD;
- (b) there exists a unital representation $\pi : \mathcal{B} \rightarrow B(\mathcal{H})$ which is not the point SOT-limit of a net of finite dimensional representations of $\mathcal{B}$.

In fact, the representation π will take a very concrete form given by

Figure 2: Hartz’s explicit algebra and Theorem 1.3, PDF pp. 4–5.

3 Idea / proof intuition

The RFD half is transparent from the concrete realization: evaluation at points of $\mathbb{T}$ gives two-dimensional representations, and the norm in $M_2(C(\mathbb{T}))$ is the supremum of the resulting matrix norms.

The obstruction appears only after taking the maximal C^* -cover. Hartz constructs the Toeplitz representation

$$\pi : \mathcal{B} \longrightarrow B(H^2 \oplus H^2), \quad \begin{bmatrix} f & 0 \\ h & \bar{g} \end{bmatrix} \longmapsto \begin{bmatrix} T_f & 0 \\ T_h & T_{\bar{g}} \end{bmatrix},$$

and proves that π is not a point-strong-operator limit of finite-dimensional representations of $\mathcal{B}$.

If $C_{\max}^*(\mathcal{B})$ were RFD, its universal property would extend π to a $*$ -representation of $C_{\max}^*(\mathcal{B})$. The Exel–Loring theorem would approximate that $*$ -representation point-strongly by finite-dimensional $*$ -representations. Restriction to $\mathcal{B}$ would then approximate π point-strongly by finite-dimensional representations, contradicting Hartz’s theorem.

Thus

$\mathcal{B} \text{ is RFD, but } C_{\max}^*(\mathcal{B}) \text{ is not RFD.}$

Hartz states this exact consequence as Corollary 1.4.

4 Provenance and scope

Hartz’s arXiv submission followed Thompson’s by sixteen months and explicitly states that the example answers the existing maximal- C^* question. The counterexample is especially sharp: the underlying algebra is unital and 2-subhomogeneous, while its C^* -envelope is the RFD algebra $M_2(C(\mathbb{T}))$; it is specifically the *maximal* C^* -cover that fails to be RFD.

This is therefore classified as a full answer already present in later literature, not as a new counterexample produced by this run.

proximated point SOT-* by finite dimensional ones; see [6, Theorem 3.3]. Indeed, necessity, which is what we will use, is immediate from the universal property of $C_{\max}^*(\mathcal{A})$ and the Exel–Loring theorem (Theorem 1.1).

Therefore, Theorem 1.3 has the following consequence.

Corollary 1.4. *Let $\mathcal{B}$ be the operator algebra of Theorem 1.3. Then $\mathcal{B}$ is RFD, but $C_{\max}^*(\mathcal{B})$ is not RFD.*

However, since $\mathcal{B}$ is 2-subhomogeneous, [2, Proposition 4.1] shows that the C^* -envelope of $\mathcal{B}$ is 2-subhomogeneous as well and hence RFD. Indeed, the C^* -envelope of $\mathcal{B}$ is $M_2(C(\mathbb{T}))$ since $\mathcal{B}$ is a Dirichlet algebra; see Remark 4.2

Figure 3: The exact negative answer in Hartz [2, Corollary 1.4], PDF p. 6.

References

- [1] I. Thompson, *Maximal C^* -covers and residual finite-dimensionality*, arXiv:2107.14275, 2021; J. Math. Anal. Appl. **514** (2022), 126277.
- [2] M. Hartz, *Finite dimensional approximations in operator algebras*, arXiv:2211.15548, 2022.